

Advanced (Habanero)

- Q1.** What is the effect of $y = f(x) + 2$ on the graph of $y = f(x)$?
(A) Horizontal shift 2 units to the left
(B) Vertical shift 2 units up
(C) Horizontal shift 2 units to the right
(D) Vertical shift 2 units down
(E) Reflection over x-axis
- Q2.** If $y = f(-x)$, what is the effect on the graph of $y = f(x)$?
(A) Reflection over y-axis
(B) Reflection over x-axis
(C) Horizontal shift 2 units to the left
(D) Vertical shift 2 units up
(E) No effect
- Q3.** What is the effect of $y = f(2x)$ on the graph of $y = f(x)$?
(A) Compression by a factor of 2
(B) Stretch by a factor of 2
(C) Horizontal shift 2 units to the left
(D) Vertical shift 2 units up
(E) Reflection over x-axis
- Q4.** If $y = 2f(x)$, what is the effect on the graph of $y = f(x)$?
(A) Vertical shift 2 units up
(B) Horizontal shift 2 units to the left
(C) Stretch by a factor of 2
(D) Compression by a factor of 2
(E) Reflection over x-axis
- Q5.** What is the effect of $y = f(x - 1)$ on the graph of $y = f(x)$?
(A) Horizontal shift 1 unit to the right
(B) Horizontal shift 1 unit to the left
(C) Vertical shift 1 unit up
(D) Vertical shift 1 unit down
(E) Reflection over y-axis
- Q6.** If $y = -f(x)$, what is the effect on the graph of $y = f(x)$?
(A) Reflection over x-axis
(B) Reflection over y-axis
(C) Horizontal shift 2 units to the left
(D) Vertical shift 2 units up
(E) No effect
- Q7.** What is the effect of $y = f\left(\frac{x}{2}\right)$ on the graph of $y = f(x)$?
(A) Compression by a factor of 2
(B) Stretch by a factor of 2
(C) Horizontal shift 2 units to the left
(D) Vertical shift 2 units up
(E) Reflection over x-axis
- Q8.** If $y = f(x + 2) - 1$, what are the effects on the graph of $y = f(x)$?
(A) Horizontal shift 2 units to the left and vertical shift 1 unit down
(B) Horizontal shift 2 units to the right and vertical shift 1 unit up
(C) Horizontal shift 2 units to the left and vertical shift 1 unit up
(D) Horizontal shift 2 units to the right and vertical shift 1 unit down
(E) Reflection over x-axis and y-axis

Open Ended Questions (FRQ)

- Q9.** Graph the function $y = 2f(x - 1) + 1$ if $f(x) = x^2$. Identify the parent function and describe the transformations.
- Q10.** Compare the graphs of $y = f(x)$, $y = f(-x)$, and $y = -f(x)$. How do the graphs of $y = f(-x)$ and $y = -f(x)$ relate to the graph of $y = f(x)$?

Chef's Tips

- Emphasize the importance of understanding function transformations.
- Use graphing tools to visualize the transformations.
- Encourage students to check their work by graphing the functions.